

4 (**short**) A quantity of air enclosed in a cylinder by a piston undergoes a fully resisted expansion from a temperature $T_1 = 280$ K and a pressure $p_1 = 4$ bar to a final pressure $p_2 = 1$ bar. The expansion can be modelled by the polytropic equation $p v^n = \text{constant}$, where $n = 1.2$. Treat air as a perfect gas with $c_v = 718 \text{ J kg}^{-1} \text{ K}^{-1}$ and $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$.

(a) Calculate the work done per kg of air. [6]

(b) Determine the final temperature T_2 and the heat transferred per kg of air. [4]

5 (**short**)

(a) Starting from the Clausius Inequality, show that the Coefficient of Performance for a cyclic refrigerator operating between two constant temperature reservoirs can be expressed as

$$COP_R \leq \frac{T_L}{T_H - T_L}$$

where T_L is the temperature of the low temperature reservoir and T_H is the temperature of the high temperature reservoir. [6]

(b) A cyclic refrigerator operates between a cold space at -10°C and the environment at 20°C . Calculate the minimum power input required to extract heat from the cold space at a rate of 0.4 kW. Why, in practice, is significantly more power required? [4]